NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR305254
Lead Applicant	Mr James Alexander
Research Title	Power-PD: EmPOWERing people with Parkinson's Disease through flexible hybrid exercise - a randomised feasibility study with pre and post-test design.
Plain English Summary	**Aim(s) of the research** I was inspired by working with people with Parkinson's disease (PD) and their feedback and evidence that group exercise can help with their symptoms. People with Parkinson's like the flexibility of doing exercise classes both in-person and online. This project will explore if Power-PD, an exercise group which allows people to attend in-person or online from home, is feasible and wanted by people with Parkinson's. This could support more people with Parkinson's to exercise regularly. **Background to the research** Parkinson's affects the body's nervous system. It causes people to feel stiff, shake and have difficulty walking and keeping their balance. It can also cause people to feel sad and anxious. Exercise is helpful, but people with Parkinson's exercise less. They say they need support to exercise regularly, especially if they have trouble walking or don't live near to an exercise group. An online exercise group could offer a way for people to exercise at home and develop exercise which can become a habit to manage their Parkinson's. Researchers say online exercise groups are safe and can help people with Parkinson's feel better. But exercising online makes it difficult to talk to and see each other. A flexible exercise group could allow people choice to attend in-person with other people or online from home. This might help people with Parkinson's exercise more. We need to see if this is true and how it might work. To answer this we have spoken to clinicians and people with Parkinson's and their families, and Power-PD has been developed. Power-PD is an exercise group for people with Parkinson's designed to offer flexible access through either attending in-person or remotely from home and develop long-term exercise habits. **Design and methods used** We will first review what research has already been done in this area and then design and deliver a study which tests whether a combined (hybrid) exercise intervention (Power-PD) is possible, and importantly if it is an intervention people want and need. *What is Power-PD?* Power-PD is for people with mild to moderate Parkinson's who do not severe difficulty with their walking and balance. Participants will attend the group 1x per week for 12 weeks. It will include 60-minutes of exercise and 30-minutes for participants to talk together about how to exercise more regularly. I will monitor the group to see how well it is carried out and ask participants for their feedback on the project. This will tell us if Power-PD should be explored as a larger clinical trial. **Patient and public involvement** I have worked with four people with Parkinson's and their family members to develop this project. They agreed on the project idea, importance of making the exercise group accessible and helped design Power-PD. We will continue working together to finalise what needs to be included in Power-PD. They will also help write easy-to-read documents for the project. **Impact & Dissemination** This study will help understand if a flexible exercise group for people with Parkinson's is needed and if so, guide a future larger study. This will provide a way for people with PD to be more physically active which is beneficial for staying healthy and managing their symptoms. The findings will be written in scientific journals. With help from Parkinson's UK charity, I will share project findings with participants from the study, other people with Parkinson's and their families and healthcare professionals through talks and information sheets. These will be co-created with people with Parkinson's.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

Table Of Contents

1. Application Summary Information
2. Applicant CV
3. Applicant Research Background
4. Plain English Summary of Research
5. Scientific Abstract
6. Detailed Research Plan
7. Patient & Public Involvement
8. Training & Development and Research Support
9. Detailed Budget
10. Uploads
11. Participants and Signatories
12. Acknowledgement, Review and Submit

1. Application Summary Information

Host organisation (which will administer any award)
University College London

Partner organisation	Central and North West London NHS Foundation Trust

Research Title	Power-PD: EmPOWERing people with Parkinson's Disease through flexible hybrid exercise - a randomised feasibility study with pre and post-test design.

Partnership Fellowship Please select the relevant Partnership Fellowship.
N/A

Proposed WTE Please select which WTE option you would prefer
1.00

Duration (months) The duration for your award is based on your chosen WTE
36

Proposed start date if grant awarded	01/04/2025

Research Type	Feasibility study

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
09/2022	07/2023	PGCHE	Healthcare research skills and methods	United Kingdom (England)	St George's, University of London	Pass
09/2011	07/2014	MSc	Advancing Spinal Cord Injuries Rehabilitation & Management	United Kingdom	Buckinghamshire New University	Pass (63%)
09/2004	07/2007	BSc(Hons)	Physiotherapy	United Kingdom	University of the West of England, Bristol	2:1

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
09/2018	-	Band 7 Physiotherapist	Central & North West London NHS Foundation Trust	0
02/2018	-	Senior Lecturer	St George's University, London	0
02/2018	09/2018	Band 6 Physiotherapist	Central & North West London NHS Foundation Trust	0
03/2015	09/2017	Teaching Fellow	King's College London	0
04/2011	12/2014	Band 6 Physiotherapist	Buckinghamshire Healthcare NHS Trust	0
12/2007	03/2011	Band 5 Physiotherapist	Salisbury NHS Foundation Trust	0
10/2007	12/2007	Band 5 Physiotherapist	United Lincolnshire Hospitals NHS Trust	0

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
HEE/NIHR Pre-doctoral clinical practitioner academic fellowship (PCAF)	I was the receiver of this personal research training award. The PCAF enabled me to undertake research skills training, develop collaborations and work with my supervisors to undertake preliminary research to support the development of a DCAF application	70758	HEE/NIHR	24	01/01/2022	31/12/2023
CNWL research starter grant	Lead investigator and applicant for the grant. Developed research protocol for study to explore	8875	Central & North West London NHS Foundation	6	03/08/2020	05/02/2021

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
	experiences of online exercise groups for people with Parkinson's disease. Used also to develop NIHR PCAF application.					
KCL/Wellcome Trust public engagement small grants scheme award	Lead applicant	750	KCL/Wellcome Trust	12	01/05/2017	30/04/2018

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
(2023) ‘Exploring people with Parkinson’s disease’s experiences of online exercise groups in the UK – a qualitative study’,
Alushi, L., Alexander, J., Jones, J. and Lafortune, L.. (2022) ‘A Systematic Review on Physical Health Education Interventions for People with Parkinson's Disease: Content, Impact, and Implementation Considerations Across the Parkinson's Trajectory.’, *Journal of Parkinson's disease*, 12(5).
Sheehan, KJ., Guerrero, EM., Tainter, D., Dial, B., Milton-Cole, R., Blair, JA., Alexander, J., Swamy, P., Kuramoto, L., Guy, P., Bettger, JP. and Sobolev, B.. (2019) ‘Prognostic factors of in-hospital complications after hip fracture surgery: a scoping review.’, *Osteoporosis international : a journal established as result of cooperation between the European Foundation for Osteoporosis and the National Osteoporosis Foundation of the USA*, 30(7).
Alexander. (2019) ‘Technology-enhanced learning in physiotherapy education: Student satisfaction and knowledge acquisition of entry-level students in the United Kingdom’, *Research in Learning Technology*, 27.
Sheehan, KJ., Williamson, L., Alexander, J., Filliter, C., Sobolev, B., Guy, P., Bearne, LM. and Sackley, C.. (2018) ‘Prognostic factors of functional outcome after hip fracture surgery: a systematic review.’, *Age and Ageing*, 47(5).
Bazin. (2016) ‘Learning with expert patients – can this support development of clinical skills?’, *Physiotherapy (United Kingdom)*, 1(102).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/07/2016	Higher Education Academy	Fellow	

ORCID iD 0000-0002-9459-7228

3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Physiotherapist

Do you have current / active professional registration? This registration must be with one of the **ICA approved regulatory bodies**
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
PH81921

Please describe your research career to date

I've established a clinical academic career working as a Senior Lecturer at St George's, University of London and a specialist neurological physiotherapist for the Camden Neurology & Stroke Service (CNSS), an NHS community neurological rehabilitation team. I have built my career by combining clinical and academic work and successfully completing my PCAF and Post-Graduate training. My ambition is to progress my clinical academic career, through completion of my DCAF, within the NHS supported by an HEI with excellent clinical and research credentials in neurology and neurorehabilitation. This will enable me to be a leading clinical academic in the field of exercise for people with Parkinson's Disease (PD), supporting and coaching others to positively impact on patients' lives. It will also enable me to impact on patient care as well as promote the role of allied health professional clinical academics through fostering national and international collaborations. I believe my research and clinical experiences to date provide strong supporting evidence for this DCAF.

Clinical experience

I have extensive experience in neurological rehabilitation, belonging to special interest groups including the Association of Chartered Physiotherapists in Neurology and the Parkinson's Excellence Network exercise hub. I have developed specialist clinical skills to support people with PD (PwP) through exercise by completing PD Warrior™ training and deepened my understanding of complex PD management through attending the Queen Square Movement Disorders short course, led by world-leading movement disorders specialists. These skills I have imparted to my and other clinical teams within CNWL NHS Trust through mentorship, teaching and clinical supervision. I will continue developing skills to become a leading clinical academic through my training and development plan. Completing the NHS Leadership Academy Edward Jenner course strengthened my leadership skills. I fostered innovation within my team and supported a service evaluation of our exercise group for people with neurological conditions alongside my physiotherapy lead. This was presented at the 2023 Congress on Neurorehabilitation and Neural Repair in Maastricht and has been submitted for journal publication.

Inspired by emerging evidence for exercise for PwP, I wanted to address gaps in service provision for this population. Embedding evidence-based research, clinical skills and utilising a Quality Improvement approach I developed an exercise and self-management group for PwP. This was achieved through logic modelling, stakeholder involvement, clinical skills training, data collection and evaluation. During the COVID-19 pandemic I transferred this to an online format. The NHS Long-term Plan, advocates greater use of e-health to provide equity of access to healthcare for those with long-term conditions and underserved populations. This initiated research questions about the role of hybrid group exercise for PwP and the need to address the gap in research. In developing the Power-PD project, logic modelling and qualitative research has supported development of my PD group. Understanding mechanisms of impact, I better embedded behaviour change theory into the group design, extending the length of the group to support this. Delivering the group in-turn has informed logic modelling of the Power-PD project.

Research experience

I have completed an MSc and a PG Certificate in Research methods. These provided excellent foundations for critically understanding and evaluating clinical research, its design, conduct, dissemination and project management and appropriate statistical methods. They, alongside GCP training, have equipped me with the knowledge of research governance and organisation to embark on a DCAF.

Research projects

I have been involved with independently designing, conducting, managing, and producing outputs from several research projects, as follows:

Clinical research:

During my PCAF I completed two studies to support development of my research question. These explored PwP's and exercise professionals' experiences of online exercise groups. The studies used a phenomenological approach through semi-structured interviews. These studies, together with post-graduate training, have given me skills in formulating a research question, methodological design, obtaining ethical approval, recruitment, data collection and analysis and publication. The findings from the study exploring PwP's experiences were presented by poster at the World Parkinson's Congress in Barcelona in July 2023, Physiotherapy UK conference in November 2023 and the SGUL Research Day in December 2023. The exercise professionals' study was presented orally at ACPIN conference 2024. Both papers have been submitted for journal publication.

During my PCAF I collaborated with a PhD student and co-investigated a systematic review into physical activity education for PwP. My role included title and abstract screening, full-text screening, data collection, quality appraisal and contributing to the reporting. This study was published in the Journal of Parkinson's Disease, a key journal in the field of PD.

I have supervised 4 undergraduate and 2 post-graduate physiotherapy student research projects. Projects have included cross-sectional and mixed methods study design and focused on physiotherapy provision for early-stage PD and student experiences of physical activity guideline promotion for neurological patients. I am currently supervising a project exploring demographic differences in perceptions of telerehabilitation in PwP.

Educational research:

I was chief investigator for a mixed methods study, utilising a survey and focus groups, exploring technology-enhanced learning in undergraduate physiotherapy education. I was responsible for study design, survey development, data collection and analysis. I presented this study orally at the World Congress of Physical Therapy and published it in Research in Learning Technology journal.

Relevant experience

I have 9 years' experience of teaching in higher education. My extensive knowledge of pedagogy and teaching methods contributed to obtaining Fellowship of the Higher Education Academy. Working within research institutes allows me to collaborate with other research-oriented AHPs and develop peer mentorship. I understand the need for clinical academics to support and promote the role within the physiotherapy profession and wider locally, nationally and internationally. I have begun to do this through my lecturing role and mentoring of staff and students. For my PGCert I focused an assignment on the role of clinical academic AHPs and have written this up as a short commentary piece that has been submitted to Physiotherapy journal. I believe I have the teaching, academic and professional skills to continue this during my DCAF.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details
A previous application to the NIHR DCAF round 3 was submitted. I was shortlisted for interview but subsequently unsuccessful. I have reflected carefully on all the feedback and I am grateful for the opportunity to revise and rethink some elements of my research, I believe this has strengthened this application.

All reviewer feedback has been addressed and I have made changes in the following key areas;

Feedback suggested that the study aims and objectives needed further clarification. These have been fully reviewed as a project team and amended to accurately reflect the aim to explore the feasibility of the Power-PD intervention and explicitly stating how this will be achieved through the objectives. The objectives have been amended to consider contribution of the scoping review to the overall project and development of the Power-PD intervention.

Feedback also suggested preference for WP2 to be designed as a randomised feasibility trial. This has been discussed as a research team. WP2 will now be a single-centre randomised (Power-PD vs usual care) feasibility study. This will enable participants' acceptance to be randomised to be established. This also strengthens the rationale for WP2 to inform future feasibility and pilot testing.

Reviewers felt that WP2 needed to include cost-benefit analysis. After discussion with the research team and PPI members, it was agreed that WP2 will now assess the feasibility of completing a cost-benefit analysis and health economic outcome. I will analyse preliminary cost-effectiveness by comparing the cost of the intervention with the cost of usual care vs resource cost (using CSRI) vs health economic cost (EQ-5D-5L & PDQ-8).

Reviewers wanted to see more consideration of hybrid exercise programmes in wider health conditions included in the rationale for the study and to inform the Power-PD intervention. Through systematic searching of Embase and Medline further literature was identified and relevant information included in the research plan background section. The inclusion criteria for WP1 will now include studies from other long-term health conditions if insufficient literature is identified in the PD population.

Consideration of who will identify adverse events, the process for communication of these to lead researcher and escalation to sponsor has been stated more clearly in the project plan.

Reviewers sought further detail on ethical implications and the impact of the study on people with PD, their carers and healthcare professionals. A meeting with PPI members explored potential concerns and these have been addressed in the project plan.

To support development of Power-PD and undertaking WP2 further training has been included in the training and development plan including "Developing and testing complex interventions" and "Design and implementation of digital health interventions."

Reviewers felt more information on the aims of the study and how these aims will be operationalised was needed in the plain English summary. Following review of the overall project aims and objectives these have been clarified and co-written with a PPI group member.

Contextual Factors

Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

Whilst i have been working in academia for 9 years my job role has primarily been focused on teaching. This has led to fewer opportunities, time and support for undertaking research opportunities with teaching responsibilities taking priority. Similarly, my clinical role offers no protected time for research and a responsibility to support clinical care taking priority. I am committed to a clinical academic career and have actively championed opportunities to embed research opportunities within my academic and clinical roles.

4. Plain English Summary of Research

Aim(s) of the research

I was inspired by working with people with Parkinson's disease (PD) and their feedback and evidence that group exercise can help with their symptoms. People with Parkinson's like the flexibility of doing exercise classes both in-person and online. This project will explore if Power-PD, an exercise group which allows people to attend in-person or online from home, is feasible and wanted by people with Parkinson's. This could support more people with Parkinson's to exercise regularly.

Background to the research

Parkinson's affects the body's nervous system. It causes people to feel stiff, shake and have difficulty walking and keeping their balance. It can also cause people to feel sad and anxious.

Exercise is helpful, but people with Parkinson's exercise less. They say they need support to exercise regularly, especially if they have trouble walking or don't live near to an exercise group. An online exercise group could offer a way for people to exercise at home and develop exercise which can become a habit to manage their Parkinson's.

Researchers say online exercise groups are safe and can help people with Parkinson's feel better. But exercising online makes it difficult to talk to and see each other. A flexible exercise group could allow people choice to attend in-person with other people or online from home. This might help people with Parkinson's exercise more. We need to see if this is true and how it might work. To answer this we have spoken to clinicians and people with Parkinson's and their families, and Power-PD has been developed. Power-PD is an exercise group for people with Parkinson's designed to offer flexible access through either attending in-person or remotely from home and develop long-term exercise habits.

Design and methods used

We will first review what research has already been done in this area and then design and deliver a study which tests whether a combined (hybrid) exercise intervention (Power-PD) is possible, and importantly if it is an intervention people want and need.

What is Power-PD?

Power-PD is for people with mild to moderate Parkinson's who do not severe difficulty with their walking and balance. Participants will attend the group 1x per week for 12 weeks. It will include 60-minutes of exercise and 30-minutes for participants to talk together about how to exercise more regularly. I will monitor the group to see how well it is carried out and ask participants for their feedback on the project. This will tell us if Power-PD should be explored as a larger clinical trial.

Patient and public involvement

I have worked with four people with Parkinson's and their family members to develop this project. They agreed on the project idea, importance of making the exercise group accessible and helped design Power-PD. We will continue working together to finalise what needs to be included in Power-PD. They will also help write easy-to-read documents for the project.

Impact & Dissemination

This study will help understand if a flexible exercise group for people with Parkinson's is needed and if so, guide a future larger study. This will provide a way for people with PD to be more physically active which is beneficial for staying healthy and managing their symptoms.

The findings will be written in scientific journals. With help from Parkinson’s UK charity, I will share project findings with participants from the study, other people with Parkinson's and their families and healthcare professionals through talks and information sheets. These will be co-created with people with Parkinson's.

5. Scientific Abstract

Background

Exercise is strongly advocated as a tool for managing Parkinson's disease (PD) symptoms with growing evidence supporting it. Due to the COVID-19 pandemic and wider accessibility issues, people with PD (PwP) have utilised online exercise groups. What is known is that PwP want both online and in-person options for group exercise. Little is known about combining an in-person and online (hybrid) format which may improve equity of access to exercise. Power-PD is an exercise group for PwP, designed to offer flexible access through attending in-person or remotely from home, and develop long-term exercise habits. We want to test whether Power-PD can help PwP sustain exercise for longer through developing these habits and integrating them into their lives, and understand the mechanisms that help them do that.

Research question

Is it possible to deliver a study exploring a hybrid exercise intervention (Power-PD) to support PwP to sustain physical activity habits?

Aim

Evaluate the feasibility and acceptability of Power-PD to support sustained physical activity habits in people with PD.

Objectives:

1. Explore the existing literature on hybrid formats of group exercise for PwP and work with them and clinical stakeholders to inform and refine the core components of Power-PD.

2. To identify knowledge and skills needed to deliver Power-PD and develop a logic model to understand and predict key mechanisms of change.

3. To determine the acceptability and fidelity of Power-PD to support sustained physical activity habits in PwP.

4. To determine the feasibility of collecting clinical, economic and qualitative data to inform future research on the cost effectiveness of Power-PD.

Methods

WP1 – To complete a scoping review exploring existing literature on hybrid interventions for PwP to refine the core components of Power-PD and logic modelling. This will follow the Joanna Briggs Institute methodology for scoping reviews and include a three-step search strategy. Data will be extracted and presented in tables and summarised narratively following critical appraisal.

WP2 – A feasibility study will assess the acceptability, fidelity, and feasibility of Power-PD. This will be a randomised, single site study with pre and post-test design using mixed methods. Participants will be randomised to Power-PD or a wait-list control. Convenience sampling will recruit participants with early-mid stage PD who will participate in a 12-week exercise and peer self-management intervention.

Timeline for delivery

Year 1: PPI activities. WP1 protocol registration and publication, data collection & analysis. Ethical approval, site set-up and training and protocol publication for WP2.

Year 2: PPI activities. WP1 outputs & dissemination. WP2 implementation, data collection, data analysis, ongoing training and support for practitioners delivering intervention.

Year 3: PPI activities. WP2 data collection, data analysis, feedback, outputs & dissemination. Thesis completion and dissemination. Future study protocol development.

Impact and dissemination

If feasible the findings will inform a definitive trial to measure effectiveness and cost effectiveness generating data for power calculations, intervention content and a refined logic model. Findings will be shared through publication in academic journals, conference presentations and talks to PwP and their families through the Parkinson's UK charity.

6. Detailed Research Plan

What is the problem being addressed?

The Parkinson's Pandemic

Parkinson's Disease (PD) is a movement disorder and the second most prevalent neurodegenerative disease after Alzheimer's disease.[1,2] It causes motor and non-motor impairments including rigidity, bradykinesia, tremor and depression.[3-5] PD is the fastest growing neurological condition in the world[5]. Its UK annual incidence is 15 to 20 per 100,000.[6]

Financial & Health-related impact of PD

Health and care costs for people with PD (PwP) are £2471 higher per patient at diagnosis increasing to £4004 after 10-years from diagnosis compared to the general population.[7] PwP are £16,500 worse off per year due to lost income from an inability to work and costs of associated disability.[8] Compared to the general population PwP are typically one-third less active, attributed to reduced physical abilities, self-efficacy and mental health impairments.[5,9,10] Disease progression and reduced physical activity (PA) worsen motor symptoms leading to falls, fractures and cardiovascular co-morbidities causing increased hospital admissions.[11] Over 4 years, 232,905 non-elective admissions in England cost the NHS over £777 million.[12] Management of PD is complex focusing upon medication who's long-term efficacy is limited inducing motor fluctuations and dyskinesias. Thus, effective long-term health interventions are required.

Why is this research important?

Exercise for PD

Exercise-related research for PwP has increased recently. Exercise encompasses activities which are planned, structured, and purposeful, with the intention of improving and/or maintaining one or more components of physical fitness.[13] Exercise is a cost-effective and low-risk intervention to improve motor and non-motor symptoms and recommended by NICE and European guidelines.[5,6,14] Due to the multiplicity of PD symptoms no one type of exercise is recommended over another and exercise can be performed individually, in a group, supervised or unsupervised.[5] A 2023 Cochrane review[15] recommended exercise for managing PD symptoms owing to its improvements in gait,[16] strength,[17] balance,[18] physical activity,[19] mood and quality of life.[20,21] PwP who are more active experience slower decline of symptoms, and improved health outcomes.[22-26] Acknowledging these benefits necessitates health interventions which support PwP to adopt long-term exercise habits. Sustaining PA is challenging. Automaticity in developing new habits takes around 10 weeks to develop.[27] Exercise interventions alone are insufficient for developing sustained PA habits. Self-management skills are integral to promote self-efficacy and behaviour change.[28] Exercise alongside self-management strategies could support self-efficacy to sustain PA.[29,30] My proposed theory of change is that small exercise habits developed in Power-PD will increase self-efficacy and initiate and maintain physical activity behaviours.

Whilst in-person exercise is popular, accessibility is difficult for people with transport or mobility issues or people in isolated communities.[31] The COVID-19 pandemic raised the profile and popularity of online delivery of exercise which is shown to be feasible among PwP.[32-36] PwP perceive online exercise lessens anxieties associated with transportation, self-confidence, environmental accessibility, and is more convenient.[37,38] However, barriers to online exercise exist, including equipment accessibility and technology illiteracy.[37,38] Despite the return to in-person healthcare provision following the COVID-19 pandemic, online exercise is still advocated for by PwP.[37]

Why is this research needed now?

The evidence gap.

Exercise for managing PD symptoms, including reduced balance and falls, is the top research priority as identified by the James Lind Alliance. Online group exercise for PwP could promote physical health benefits but this has not yet been supported in clinical trials.[39-42] Programmes within NHS services that illicit and sustain PA habits in PwP are needed, helping to support standardisation of care provision of exercise.[39] During my PCAF I undertook research demonstrating PwP want a hybrid approach to group exercise offering flexibility to attend online and/or in-person.[37] This is also supported by my PPI group, professional advisory group and other research.[38] Hybrid exercise could improve equity of access, allowing PwP to sustain PA habits. This is underpinned by behaviour change theory that through increasing opportunity and flexibility to exercise helps to increase participation.[43] The use of technology to support adherence to exercise programmes is recommended.[44] Therefore, the rationale for a

hybrid format allowing flexibility for participants to attend online or in-person is strong. Hybrid exercise groups which offered in-person group exercise followed by remotely supported independent exercise in the community have shown good feasibility and acceptability in PD,[36] cardiac rehabilitation,[45,46] and oncology[47] patients. No studies on hybrid group exercise interventions (attending either in-person or online) have been identified in PD or other long-term conditions. Power-PD will be the first to explore this. A hybrid format of pulmonary rehabilitation within CNWL NHS trust has demonstrated this approach to be feasible for both patients and healthcare professionals. What is unknown is whether a feasibility study exploring a hybrid format of group exercise is feasible or acceptable to PwP.

What is the research question / aims and objectives?

Research question

Is it possible to deliver a feasibility study exploring a hybrid exercise intervention (Power-PD) to support PwP to sustain physical activity habits?

Aims and objectives.

Evaluate the feasibility and acceptability of conducting Power-PD as hybrid intervention to support sustained physical activity habits in people with PD.

PPI helped refine the research aims and objectives. This project utilises MRC guidance on developing and evaluating complex interventions and builds on developmental work completed during my PCAF.[48] Objectives were identified to continue the development of this complex intervention in preparation for future feasibility and pilot testing.

Objectives:

1. Explore the existing literature on hybrid formats of group exercise for people with PD and work with PwP and clinical stakeholders to inform the core components of Power-PD.

2. To identify knowledge and skills needed to deliver the Power-PD intervention and develop a logic model to guide its operation.

3. To determine the acceptability and fidelity of the Power-PD intervention to support sustained physical activity habits in PwP.

4. To determine the feasibility of collecting appropriate quantitative, qualitative, clinical and economic outcome data to allow further formal assessment of Power-PD.

Project Plan

Work package 1 (WP1) – Scoping review

A scoping review is chosen to map key concepts underpinning the topic and clarify working definitions, and the conceptual boundaries.[49] A paucity of published studies exploring hybrid exercise groups for PwP elicited discussions with the PPI group and project supervisors and agreement for a scoping review as the preferred methodology at present.

<u>Aim:</u>

To evaluate the existing evidence base supporting hybrid formats of group exercise for PwP to inform Power-PD intervention design and operation.

<u>Objectives:</u>

This will inform WP2 in two ways:

1. Support development of a theory of change through exploring the key components of other interventions, the mechanisms of these interventions, their interactions and influences on outcomes to inform the core components of Power-PD.
2. Identify knowledge and skills needed to deliver the Power-PD intervention and develop a logic model to guide its operation.

The objectives will be achieved by answering the following questions:

- What are the operational definitions used for online and hybrid exercise group studies?
- What study methods have been used to investigate online and hybrid exercise group interventions?
- What are the common intervention components of online and hybrid exercise groups?
- What outcome measures are used to assess online and hybrid exercise groups?

Participants

Included studies will target people with idiopathic PD at all stages (Hoehn & Yahr stages 1-5) of all ages. Studies specifically looking at people with atypical PD will be excluded as exercise approaches for these conditions differ compared with idiopathic PD.

Concept

Due to the emerging nature and limited published literature on hybrid formats of group exercise the review will include all formats of online group exercise, including hybrid formats specifically for PwP. If insufficient studies are identified specifically for PwP then studies in other long-term conditions will be considered. A group is defined as an intervention involving two or more participants. The intervention can include any form of exercise/PA. Included studies will need to have delivered a “live online” exercise group i.e., not solely pre-recorded, and deliver an exercise/PA component with or without other components e.g., education or peer support.

Context

Studies will be included if the intervention took place online e.g., online video conferencing platforms, other digital platforms and mobile apps. Studies which have incorporated a hybrid approach will be included.

Types of Sources

This review will consider experimental and quasi-experimental study designs including RCTs, non-randomized controlled trials, before and after studies, interrupted time-series studies, piloting and feasibility studies. Due to the novel nature of eHealth interventions, studies without control groups will be included. Additionally, analytical observational studies including prospective and retrospective cohort studies, case-control studies and analytical cross-sectional studies will be considered for inclusion. We will consider descriptive observational study designs including case series, individual case reports and descriptive cross-sectional studies for inclusion. Qualitative studies will be considered that focus on, but not limited to, designs such as phenomenology, grounded theory, ethnography, qualitative description, and action research. Additionally, systematic reviews that meet the inclusion criteria will be considered, depending on the research question. Finally, text and opinion papers will be considered. Only studies published in English will be included. There will be no limitation on publication date.

Methods

This review will be conducted in accordance with the Joanna Briggs Institute (JBI) methodology for scoping reviews.[50]

Search strategy

The search strategy utilises the JBI 3 step methodology. It will identify published and unpublished studies. An initial limited search of MEDLINE, CINAHL and Embase will identify articles on the topic. This preliminary search will use these selected keywords: (Parkinson’s disease or Parkinson’s) AND ((online exercise) OR (virtual exercise) OR (remote exercise) OR (e-health) OR (home-based exercise) OR (telehealth) OR (telemedicine)) AND ((Exercise OR (physical activity)). Words contained in the titles and abstracts of relevant articles and the index terms used to describe the articles will aid development of a full search strategy for selected databases: MEDLINE/PubMed, CINAHL, Web of Science, Embase, Cochrane Library and PsychInfo. The search strategy will be adapted for each included database and/or information source. The search strategy will be an iterative process

and may be adapted upon familiarity with the evidence base.[50] Reference lists of included sources will be screened for additional studies. Sources of unpublished studies/ grey literature to be searched include Google, Web of Science, Embase and Clinical Trials Register.

Study/Source of Evidence selection

All identified citations will be collated, uploaded into EndNote and duplicates removed. Following a pilot test, titles and abstracts will be screened by two or more independent reviewers against the inclusion criteria. Potentially relevant sources will be retrieved in full, and their citations imported into EndNote. Full texts of selected citations will be assessed against the inclusion criteria by two or more independent reviewers. Reasons for exclusion at full text will be recorded and reported in the scoping review. Disagreements between reviewers during the selection process will be resolved through discussion or with an additional reviewer/s.

Data Extraction

Data will be extracted by two or more independent reviewers using a data extraction tool developed by the reviewers, informed by JBI data extraction recommendations[51] but modified to include all pertinent information relevant to the review aims and objectives. The tool will be pilot tested by two researchers by extracting data from 3 papers and comparing results. The data extraction tool will be modified as necessary during data extraction.

Data Analysis and Presentation

Data will be presented in a table summarising the evidence. A narrative summary will accompany the results. Categories for data presentation include: year of publication, author(s), country, setting, population and sample size, research design and methods, intervention type, outcome measures, and impact and results. A lay summary will be co-produced with the PPI group.

Work package 2 – Feasibility study

Aim:

To conduct a mixed methods feasibility study to explore the acceptability, feasibility and fidelity to inform future research on the cost effectiveness of Power-PD.

Objectives:

- identify ways to ensure representative recruitment, engagement and acceptability to being randomised
- monitor adherence to the intervention
- clarify the time needed to collect, clean and analyse data
- evaluate the practicality of delivering the intervention in the proposed setting
- explore the feasibility of assessing cost effectiveness of the intervention
- explore the signal of efficacy of intervention
- confirm the choice of clinical outcome measures

Methodology

Study design

A feasibility study will be done to ascertain that this novel intervention can be delivered as an RCT.[52] It will be a single centre randomised (Power-PD vs wait list) study with pre and post-test design (See figure 1). This allows intervention and waitlist group comparisons and pre and post intervention comparisons (within subjects). A waitlist group enables all participants to gain the potential benefits of the intervention.[53]

Sampling

To achieve a sample size at the upper end of target it is estimated 150 eligible participants will need to be screened accounting for a 35% participation rate to within a 95% confidence interval of +/- 10%.[54] This is based on a recent pilot study of PD participants who reported a 35% participation rate and 11% attrition rate.[55]

Inclusion criteria:

- Idiopathic PD

- Stable medication for >3/52
- Hoehn & Yahr (PD rating scale) I-III
- Mild to severe gait disturbance with a score of ≤2 on the Movement Disorders Society-Unified Parkinson's disease Rating Scale (MDS-UPDRS) item 29
- Aged 18+
- Safe to do physical exercise at home.

Exclusion criteria:

- Atypical Parkinsonism
- Cognitive impairment (MMSE score of <21/30)
- Unable to walk independently 10 m without aids or assistance.
- Severe, unpredictable motor fluctuations
- Comorbidities preventing safe exercise e.g., heart or lung conditions, uncontrolled diabetes, other neurological disorders, cancer or musculoskeletal conditions e.g., back pain or severe arthritis.

Recruitment will be carried out in partnership with NHS general practices, neurology clinics and therapy services in a North London borough. They will identify eligible PwP who meet inclusion criteria and provide them with a participant information sheet. Interested participants will register their interest via telephone or email. Screening and eligibility checking will be conducted by the principal investigator before obtaining informed consent.

Power-PD may offer greater inclusivity by allowing flexibility in participants mode of attendance. To support recruitment of PwP from Black, Asian and Mixed Ethnicities (BAME), often under-represented in research, I will collaborate with general practices with higher BAME populations as recruitment sites and work with the local Parkinson's UK support group to identify a trusted advocate to support accessing local communities. To support recruitment from lower socioeconomic groups, keen to participate but lack access to IT equipment or internet, tablets and dongles will be provided for the duration of their intervention.

Intervention

Power-PD has been developed through an iterative process combining theory, evidence-based clinical practice, and collaboration with my PPI, advisory group and clinical team to increase the likelihood of behaviour change and intervention implementation.[56,57] Intervention mapping, logic modelling, theories of implementation science and the COM-B model of behaviour change were used to develop the intervention.[58] The intervention will be refined with input from academic and clinical supervisors, PPI, stakeholders and findings from WP1.

Power-PD is a 90-minute exercise and peer self-management group over 12 weeks designed to build exercise self-efficacy and support behaviour change. Participants have flexibility to attend in-person or virtually through Microsoft Teams and can switch between formats each week depending on individual needs. Microsoft Teams has been successfully used previously with PwP when running an online exercise group by myself. A maximum of 10 participants per group aligns with best practice guidelines regarding health & safety within the gym space and for online management. A home risk assessment will be completed with participants, to advise on suitable environments for exercise and how to use the MS Teams platform, following guidance from Parkinson's UK and the CIMSPA policy for 'safe delivery of exercise at home.'[59,60]

Participants will receive a booklet containing tools to support them during the intervention including outcome measures, PA diary, exercise guides, and resources to facilitate exercise self-management e.g., education, reflective diaries, goal-setting and problem-solving worksheets. PPI and stakeholders helped inform the exercise content which will be refined and finalised upon findings from WP1. The exercise component will last 60 minutes (10-minute warm up, 40-minute circuit-based exercise and 10-minute cool down). Exercises will encompass strength, aerobic, balance, cognitive, gait, and functional components, in dual and single task activities as per guidelines.[5] Participants will be encouraged to work at a moderate-high intensity (60-80% rating of perceived effort). Recordings of the exercises will be emailed to participants supporting independent exercise. The class allows online and in-person participants to exercise together. A physiotherapist and therapy assistant will deliver the class to in-person participants and the online participants. Instructors will provide feedback, refine technique and tailor exercises to participants as appropriate. The physiotherapist will have experience in neurology and delivering an exercise class. The therapy assistant will have experience of supporting an exercise class. Training will be provided reflecting the core components of Power-PD and the skills needed to deliver its approach.

Following the exercise component, a 30-minute peer self-management programme devised in collaboration with PPI members, will be facilitated by the instructors. Peer learning has been shown to enhance self-efficacy through shared problem-solving, activity habits and successes which may support sustained PA habits.[61] Self-management programmes embedded in, behaviour change theory, self-efficacy principles and devised with PPI are more successful in their implementation.[62] Topics included in Power-PD are based on established

interventions including reflection, self-discovery, self-monitoring, problem-solving and activity and goal setting to support physical behaviours and cognitive process.[63,64] Content will be refined during year 1 of the DCAF through an iterative approach in collaboration with PPI and academic supervisors.

Data collection

A mixed methods approach will measure the acceptability, feasibility and fidelity of Power-PD. This approach will enable triangulation, completeness and explanation when considering feasibility outcomes. Participant burden was considered by the PPI group when finalising the number of outcome measures. Outcome measures related to the study aims and objectives are in Table 1.

Primary outcomes:

Acceptability: Demographic data including age, socioeconomic status, education level and ethnicity will be collected to review representative engagement. Prospective acceptability using the Theoretical Framework of Acceptability (TFA) questionnaire will explore eligible participants' reasons for non-participation or dropping out and willingness to be randomised.[65,66] The TFA enables assessment of acceptability in the design, evaluation, and implementation of healthcare interventions.[66] The questionnaire will contain both open and closed questions. All participants who complete the intervention and practitioners delivering the intervention will receive the questionnaire and be invited for a semi-structured interview to explore acceptability and experiences of the intervention. Interviews will be conducted by the principal investigator utilising a topic guide using the TFA.[66] Interviews will be conducted at a location convenient to the participant and researcher, recorded via digital voice recorder and fully transcribed. A fieldwork journal will be kept when undertaking interviews to note additional relevant details which arise. This data will add context to findings as a form of triangulation. Transcripts will be sent to interviewees to confirm accuracy.

Feasibility: Will be assessed by collecting data on recruitment, adherence, completion rates and adverse events. Recruitment rate will be derived from eligibility and consent rate. Adherence will be derived from the number of attended sessions per participant. Completion rate will be derived from retention of participants from start to primary end point and the number of completed datasets. Adverse events during the intervention period will be reported in a log and provided to the Sponsor. Severe adverse events, including those leading to hospitalisation (e.g., falls), life threatening events (e.g., cardiac events) and death will be identified and noted by practitioners delivering Power-PD, discussed with the project lead and reported immediately to the Sponsor using a reporting form and the study paused until considered safe to continue or terminated.

Fidelity: Will be assessed through participant and practitioner interviews and direct observation of six group exercise sessions by the principal investigator to establish whether the intervention is delivered as planned against a checklist of core intervention components.[67] Fieldnotes will be kept by the observer which will be used to add context and explanation to findings as a form of triangulation.

Secondary outcomes:

The choice of secondary outcomes were influenced by the PPI group. Signal of efficacy will be assessed through effect size of secondary outcomes measured in-person at baseline & immediately post-intervention. This will provide preliminary data to inform sample size calculations for future pilot/feasibility testing.

Physical activity outcomes

Physical activity diaries: PA diaries are frequently used to record and measure levels of PA in health population research due to their practicality, low cost, low participant burden, and general acceptance.

Body-worn accelerometer: The GeneActiv™ monitor from Active Insights is a body worn triaxial accelerometer that monitors PA and activity intensity (Activinsights.com). Triaxial accelerometers have been shown to accurately measure PA in PwP and are tolerated well by participants.[68] The accelerometer will be worn during the period of the intervention.

Functional mobility outcomes

Timed Up & Go (TUG): One of the most used and reported measures for functional mobility and evaluating intervention effects in ambulatory PwP. It measures the time needed to rise from an armchair, walk three meters, turn, walk back and sit down again. The TUG has demonstrated satisfactory clinimetric properties in PwP.[69] It has also shown ability to discriminate between early and middle disease stages.

Self-efficacy outcomes

Self-Efficacy for Exercise scale (SEE): Based on work by McAuley's 1990 (unpublished)[70] self-efficacy barriers to exercise measure. It's a 9-item self-rating instrument measuring self-efficacy through belief in the ability to continue exercising in the face of barriers in sedentary adults in the community. There is statistically significant correlation between efficacy expectations related to exercise and adherence to a regular exercise program.[71] There is evidence for the reliability and validity of the SEE scale for use with diverse populations.[72]

Importance/Confidence measure: To support health outcomes, measures of patients' readiness to change is recommended in clinical practice.[73] One aspect of readiness is the confidence in the individual's ability to succeed and importance placed on changing.[73] Both are linked to self-efficacy theory. Single measure visual analogue scales can be used to measure both importance and confidence to self-manage PA.[73]

Cost benefit analysis

A preliminary analysis of cost-effectiveness will be undertaken. The EQ-5D-5L and PDQ-8, a specific measure of quality of life in PWP, will capture health economic outcomes and the Client Service Receipt Inventory will collect information on resource use. Further training in health economics in clinical trials has been identified in the training and development plan to support undertaking cost-benefit analysis.

Data analysis

Interview data will be coded using NVivo software[74] following a framework analysis approach to aid triangulation of findings with the TFA questionnaire and field notes. An inductive and deductive approach will develop a coding framework based on the 8 TFA constructs. Coding and categorisation of data will be reviewed by a second researcher. Individual participant themes derived from data analysis will be sent to those participants for corroboration.

Demographic data will be presented descriptively including means (with standard deviation) for continuous data and raw count (percentages) for categorical data. Primary and secondary outcomes will be analysed using simple descriptive statistics. Findings will be presented as medians (range) for ordinal data, mean (95% confidence interval) for continuous data and raw count (number, percentage) for nominal data. A Cohen d test will measure effect size between secondary outcomes at T0 (baseline) and T1 (12 weeks) timepoints. All data analysis will be completed using SPSS.[75]

Quantitative and qualitative findings will enable conclusions and inferences about further feasibility testing of Power-PD and a future RCT.[76] Triangulation will be used to analyse acceptability of the study through questionnaire acceptability ratings and qualitative data from the questionnaire open-ended questions and semi-structured interviews. Integrative analysis using a comparison joint display regarding acceptability will be employed.[77] To analyse recruitment, data on recruitment rate and individual rationale for study participation from interviews will provide completeness of data. Adherence will be analysed through completion rate, attrition rate, field notes and open-ended questionnaire responses from participants who drop out of the study. Integrative analysis using a synthesis joint display with the complementary quantitative and qualitative data findings will address this specific domain.[77] Fidelity outcomes will include quantitative data on the fidelity measure checklist alongside field notes and participants' and professionals' interview data. This will provide explanation for any differences in participant or professionals' characteristics and context on barriers or facilitators to delivering the study. Integrative analysis using an interconnection joint display will uncover differential patterns within this domain.[77]

Progression criteria

A traffic light system will establish potential progression to a further feasibility testing (≥80% proceed, ≥60 - ≤79% proceed with rescue package, ≤59% not proceed and return to development and refining the intervention design). The progression criteria to be used are the following: (a) participant recruitment, (b) mean participant adherence, (c) completed data sets, (d) participant and instructor acceptability and (e)consideration of adverse events.

Dissemination, Outputs and anticipated Impact

What do you intend to produce from your research?

I will register and publish study protocols for WP1 & 2. Following completion of both WPs I will produce peer-reviewed publications. To reach wider audiences I will present the findings of both WPs at relevant conferences. If WP2 findings meet progression criteria I will produce a protocol for a future pilot/feasibility study.

How will you inform and engage service users, services and the wider population about your work?

World Parkinson's Congress (WPC) brings together academics, clinicians and PwP. I will present findings from my scoping review at the WPC 2026 conference. To inform PwP, who have less access to scientific papers and conferences, I will feedback through webinars with Parkinson's UK charity, presentation at a Parkinson's UK support group within the recruitment area during years 2 & 3 of my DCAF. I will work with my PPI group during times when outputs and dissemination are occurring to ensure content is appropriate to PwP and the wider population.

How will your outputs enter our health and social care system or society as a whole?

I will produce accessible infographic posters and guidance leaflets for healthcare professionals and PwP. These will be disseminated through my professional social media (X), academic platform (ResearchGate) and through the Parkinson's UK exercise Hub. I will present WP findings at a UCL Partners Centre for Neurorehabilitation seminar. Through collaborations with the Parkinson's Excellence Network and Professor Bas Bloem I hope to influence policy and guidance on exercise provision for PwP.

What further funding or support will be required if this research is successful?

I will seek further NIHR or charitable funding for post-doctoral study to complete a systematic review to further explore online exercise groups, and to conduct further feasibility testing in preparation for a future RCT.

What are the possible barriers for further research, development, adoption and implementation?

A lack of research funding and/or clinical academic career positions to enable continued clinical research and career development. Changes to current licenses for NHS staff to use Microsoft Teams software. Lack of support including commissioning of hybrid exercise groups could be a barrier to uptake of findings into the health and care services.

What do you think the impact of your research will be and for whom?

Good research needs to be built. Findings will inform if a definitive trial to measure effectiveness and cost effectiveness is feasible and inform power calculations, intervention content and refine the logic model**.**

Project Management

I will undertake managing the DCAF with academic and clinical support from my supervisors building on the strength and success of our team during my PCAF. Monthly meetings with my primary supervisor and bimonthly meetings with my both academic supervisors will provide ongoing guidance during the DCAF. Adherence to my GANTT timeline and quarterly meetings with the PPI and advisory groups will provide guidance and hold accountability to manage the project to time. An element of slack has been included in timings to mitigate for potential delays. I will use a weekly journal to specify activities to be completed and break down larger chunks of work into smaller work packages. This will be reviewed at each monthly supervisor meeting. I will use Microsoft Project software for this.

The financial management of the award will be managed through the research and departmental administrators of the Department of Clinical and Movement Neurosciences, UCL.

Ethics

WP1: Ethical approval not required. No ethical concerns identified.

WP2: Ethical approval will be sought prior to commencement of the study through the Health Research Authority through the Integrated Research Application System. Participants will provide signed informed consent. Participation or not will not affect input from NHS care services. Data will be stored in a password protected online server (REDcap or OneDrive) and comply with GDPR guidelines. A Risk assessment will consider the study risks. Participants will have a process to raise comments and complaints that may arise.

The PPI group were utilised to consider ethical considerations of the project. The following ethical concerns were raised and have been considered in the project plan; equity of experience for those participating online as in-person, ability of participants to receive tailored exercises to suit their individual needs, clear advice and guidance for family members and carers to support participants to engage with the study. Financial incentives were also considered important by PPI members for participants as a gesture of appreciation. This has been factored into costing of the study.

Success Criteria

Measurements of success

- Registration and publication of protocols for WP1 & 2
- Completion of both work packages
- Publication of WP1 & 2
- Completion of training and development programme
- Successful completion of dissertation and defence of PhD

Key risks to delivering this research.

Risks	Contingencies
Unable to recruit sufficient participants and staff to WP2	• Work with North Thames CRN. • Utilise existing knowledge and contacts from recruitment area. • Utilise support from supervisors who are clinicians working in NHS sites in the recruitment area. • Costing for incentives to participation and cost of attendance.
Training & Development courses not running as planned	• I will work closely with academic and clinical experts to identify suitable alternatives should planned training be cancelled. This could include academic training provided by UCL. I will also seek relevant academic support from within my supervisory structure.
Delay in IRAS ethical approval for WP2	• Utilise SoECAT and feasibility study protocol to support IRAS application. • Prompt ethical application to IRAS to provide sufficient time for any amendments.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

During my PCAF I collaborated with Parkinson's UK and established an advisory group for the development of the DCAF application and research proposals. The advisory group consists of 4 people with Parkinson's disease and one family member/carer from all over the UK. They were considered appropriate to involve due to their relevant and direct experience of PD. It's based on mutual respect and shared values on the role of exercise for PD. There have been 4 online meetings focusing on different aspects of the research proposal, work packages and DCAF application. To support effective PPI I undertook training from the NIHR of how to conduct PPI: focus groups and more.

Meeting 1: Focused on exploring the topic idea, research question, aims and objectives. Discussions focused my attention to consider equal opportunity and how to measure accessibility to Power-PD, including participants from lower socioeconomic backgrounds or those with low exercise self-efficacy. From this I refined the recruitment process for the WP1.

Meeting 2: Contributed to logic modelling through identification of what to include in the Power-PD exercise intervention. This enabled me to consider exercise type, class structure and equipment required to include in costing of the study. We also discussed what outcome measures were important and meaningful to assess. This enabled me to select secondary outcome measures.

Meeting 3: This focused on feeding back to participants on the DCAF aims and WPs 1 & 2 to collaboratively create the intervention name. Discussions on barriers to self-managed physical activity contributed to development of the self-management programme for Power-PD.

Meeting 4: Focused on ethical concerns with WP2. PPI members identified ethical concerns regarding equity in care provided between those attending online and in-person. The acceptability of health economic measures including EQ-5D-5L and PDQ-8. were discussed and PPI members reviewed the measures and provided comment on how easy these outcomes were to complete.

Two members of the PPI group also supported me with writing of the plain English summary for the DCAF application.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

During my DCAF I will continue working with the PPI group, expand the group to more members through support from Parkinson's UK. The aim is to include PwP who are keen to exercise but struggle to do so or are less confident with online formats of exercise. This will help to introduce insights and perspectives to refine WP2 acceptability to PwP and recruitment strategies. Due to the length of the project and unpredictability of PD, membership of the PPI group will be rolling to allow flexibility to participate. The PPI group will continue to meet 3x yearly online. Meetings will have pre-set agendas which will be forwarded to participants prior. Meetings will be recorded so information can be reviewed and revisited at later time points. All people involved will be appropriately recompensed for their time and provided with training they need, in accordance with NIHR PPI guidance.

The aim of early PPI input is to consider recruitment strategies for WP2 including language use in advertising and participant information sheets. It will consider how to identify and encourage recruitment of PwP from lower socioeconomic backgrounds and those with low exercise self-efficacy. PPI input will also be to continue to refine the intervention content (exercise and self-management programme) of WP2. This will be considered alongside findings from WP1.

Additionally, PPI input will be used to support study documentation. This includes reviewing the language and content of the participant booklet for WP2. There will also be input to aid completion of the ethics application for WP2 including writing of the plain English summary and participant information sheet.

Finally, PPI will help with dissemination of the findings from WP1 and WP2 through review of summary documents and infographics prior to being sent to participants or for public dissemination.

I have established an advisory group of academics and healthcare professionals. Although this is not the same as PPI, this will allow for perspectives from clinical academics and healthcare professionals which may assist with supporting acceptability of the feasibility study to healthcare professionals and translation of the study into clinical practice.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
N/A

8. Training & Development and Research Support

Proposed training and development programme

Academic training and development

I have utilised the Vitae research development framework (vRDF) to identify training to build on my existing skills and capabilities to continue developing as a clinical academic and support my proposed research.

The UCL Doctoral Skills Development Programme (DSDP) provides free training on topics embedded within the vRDF to develop skills for research, professional development, and employment. I will strategically organise my time over the 3 years to best support my research needs and leadership development and attend the following training (not exhaustive):

- Knowledge & intellectual abilities: Creativity in research; Introduction to EndNote; Statistics for researchers.
- Personal effectiveness: Influencing & negotiating; Managing as a researcher; Producing thesis and reports.
- Research governance & organisation: Transparency in research, Grant and charity funding.
- Engagement, influence & impact: Writing style and process; Planning for impact.

The UCL Queen Square Institute of Neurology (IoN) has some of the world's most influential academics across disciplines in neurosciences and neurorehabilitation. It supports academic development through research groups, seminar series and informal peer-support. PhD students have access to an induction programme and buddy scheme. Its strong clinical partnership with the National Hospital for Neurology and Neurosurgery and Royal Free Hospital offers high quality neuroscience teaching through integration of research, clinical practice and education.

Academic courses relevant to research project

- Joanna Briggs Institute (JBI) scoping review 1-day workshop. To enable exploration of the theories and concepts relating to scoping reviews and other types of evidence synthesis and equip myself with the knowledge and tools required to undertake and report WP1.
- Pilot and Feasibility Studies 1-day workshop (Queen Mary, UoL). To enable greater understanding in the planning and reporting of pilot or feasibility studies. This will support planning, undertaking and reporting on WP2.
- Developing and testing complex interventions (Future Learn). To design and deliver randomised controlled trials of healthcare interventions to maximise their effectiveness. This will aid WP2 development.
- Design and implementation of digital health interventions (Imperial College London). To gain understanding of design, regulatory approaches, ethics, technology adoption, implementation and strategy as applied to digital health. This will aid WP2 development.
- Understanding health economics in clinical trials (UCL). To understand facilitating a high-quality economic evaluation in a clinical trial underpinning the purpose of economic evaluation to provide evidence for policy makers. This will support feasibility assessment of cost-benefit analysis in WP2.
- Randomised controlled trial course, 5 days in-person (Oxford University). To provide grounding in the principles and practice of RCTs for evaluating healthcare interventions. This will support development of a protocol for post-doctoral research from completion of WP2.
- Emerging Clinical Leaders. 6 days virtual attendance over 6 months (King's Fund). To develop skills to succeed in the modern health and care environment, collaborating with colleagues from other disciplines and build my career as a clinical academic.

Personal and professional practitioner training and development:

I will undertake the following clinical training and attend these relevant conferences to support my development as a world-leading clinician in the field of PD. The courses will allow me to teach my clinical colleagues best practice for this client group also.

- Parkinson's Disease and Movement Disorder Course, University of Salford. A 30-credit blended distant learning module over 1 semester to enable myself to systematically and critically apply theory and practice in the management and care of people with movement disorders acting with clinical autonomy. It requires 3 days on-campus attendance.
- PWR!Moves Therapist Certification 2-day in-person Workshop, Tuscon, Arizona. To master this emerging evidence-based therapeutic concept which utilises progressive exercise and the PWR!Moves for PD-specific functional skill training across a range of disease severity and real-world environments.
- Atypical Parkinson's 1 day in-person course (NCORE). To gain an increased knowledge and understanding of these conditions including therapy management and person-centred care.
- Parkinson's: Train the trainer Course (Parkinson's UK, 3hrs virtual attendance). A course for experienced

health professionals to become a Parkinson's Excellence Network trained trainer & learn how to deliver PD training to their colleagues.

Conferences

- Year 2: Association of Chartered Physiotherapists in Neurology conference, 2026, UK. An opportunity to disseminate findings from WP1 at a national conference and network for post-doctoral collaborations.
- Year 2: World Parkinson's Congress (WPC) 2026, Phoenix, USA. An opportunity to network with clinical specialists in PD and present findings from WP1.
- Year 3: World Congress of Physical Therapy (WCPT) 2027. TBC. An opportunity to disseminate preliminary findings from WP2 at an international conference and network for post-doctoral collaborations.

Clinical practice to be undertaken

Further to my training, I will work in varied clinical practice roles to maximise my understanding of healthcare management for PwP throughout the care pathway. This will include working with NHS colleagues in neurology outpatient clinics and community services and with private practice colleagues supporting PwP through online exercise groups.

Year 1: Camden Neurology & Stroke Service (CNSS) (0.2fte). I will continue to run community-based PD exercise groups. Working with CNSS will develop collaborations to recruit to WP2.

Year 2: Neuro Heroes (0.2fte). A private online company providing 1:1 and group-based exercise for people with PD and other neurological conditions. Working with Neuro Heroes will develop my organisational, management and clinical skills in delivering online exercise.

Year 3: National Hospital for Neurology & Neurosurgery (NHNN) (0.2fte). Working in a neurological outpatient department run by specialist neurologists, nurses and physiotherapists will develop clinical skills to support PwP recently diagnosed and at different stages of their disease compared to the CNSS. Working with NHNN will develop collaborations to recruit to WP2.

Timetable

	Conferences	Academic training		Clinical training	Clinical practice
Year 1		UCL DSDP training	JBI scoping review workshop Queen Mary UoL, pilot & feasibility studies workshop Developing and testing complex interventions Design and implementation of digital health interventions Understanding health economics in clinical trials (UCL)	PD and Movement Disorders Course, Uni of Salford	CNSS
Year 2	ACPIN & WPC		Oxford Uni, RCT course	Atypical Parkinson's Course, NCORE PWR!Moves workshop, Tuscon, Arizona	Neuro Heroes
Year 3	WCPT		King's Fund, Emerging clinical leaders course	Parkinson's: Train the trainer Course, Parkinson's UK	NHNN

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Professor Anette Schrag

Institution	University College London

Position	Professor of Clinical Neurosciences

Description of support provided
Professor Schrag is a professor of clinical neurosciences within the department of clinical and movement neurosciences at the UCL/Queen Square Institute of Neurology. Professor Schrag researches clinical aspects of PD and other movement disorders. Her research has focused on patient-centred healthcare, with work concentrating on patient-reported outcomes and quality of life. She received the Silver Medal of the European Society for Person Centred Healthcare in 2018 related to this work. She is currently running several clinical trials as Chief Investigator in the NHS including PD-Care (Development of a facilitated Self-Management Tool for patients with Parkinson's). Prof Schrag has supervised nine PhD students, five of whom as primary supervisor (three completed, two ongoing), and four as secondary supervisor (two completed, two ongoing), one medical doctorate student (completed) and 27 Master students (21 completed, six ongoing). Prof Schrag and I share an interest in exercise and self-management interventions for PD. We have already established a constructive working relationship during my PCAF and she is keen to continue as my primary supervisor during my DCAF. We have maintained virtual meetings every 2 months and plan to meet monthly during my doctoral studies. She has experience of a wide variety of research methodologies including scoping reviews and clinical trials, which will be valuable for supporting my research plan. Prof Schrag has an extensive network of peers and colleagues nationally and internationally, including Prof Bas Bloem, which will be of substantial benefit for developing collaborations with world-renowned clinical and academic experts in this research area.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Professor Fiona Jones

Institution	St George's, University of London

Position	Professor of Rehabilitation Research

Description of support provided
Professor Fiona Jones is Professor of Rehabilitation Research at St George's University of London and the founder and CEO of 'Bridges self-management' which works alongside people with long-term conditions to design and deliver best practice in self-management. Her research interests lie in neurorehabilitation and self-management specifically inclusive approaches to self-management support and sustainability within rehabilitation settings. She is leading and collaborating on several research projects related to personalised self-management support including the NIHR funded Listen project (Long COVID Personalised Self-management support- co-design and Evaluation). Fiona has supervised 8 PhD students to completion. She has also experience in PhD examination; upgrade MPhil to PhD (3 students) and PhD viva (13 students). We share interests in clinical neurological research and although her work focuses predominantly on stroke it is easily transferrable across different neurological populations. Prof Jones's has supported me, alongside Prof Schrag, as my secondary academic supervisor during my PCAF. We have met as a research team every 2 months and Prof Jones has provided valuable support for the development of the two work packages, especially with guidance on logic modelling to develop the feasibility study. Meetings will continue during my DCAF. Prof Jones' expertise in self-management and her experience of designing interventions for addressing physical inactivity will undoubtedly be of value during my doctoral studies. Prof Jones will also provide support to help develop my route to achieving high research impact and skills to communicate and disseminate my ideas and research findings to different stakeholders.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Dr Caroline Appel

Institution	Central & North West London NHS Foundation Trust

Position	Lead Physiotherapist Camden Neurology & Stroke Service

Description of support provided
Dr Caroline Appel is the clinical lead physiotherapist within the Camden Neurology & Stroke Service (CNSS), an NHS community neurological rehabilitation team. Caroline has been working as a clinical academic since she completed her PhD in 2015. Her thesis explored shoulder strapping for stroke-related upper limb impairment. Caroline understands the difficulties in developing a clinical academic career in the NHS and barriers to conducting clinical research in the NHS. Caroline and I have worked together for over 6 years, and we have an excellent professional relationship. We have been working together to build research capacity into the CNSS which Caroline has taken a lead on. Caroline has been supportive of my clinical academic career and supported me in successfully applying for grants. She has also regularly attended PCAF supervisory meetings to contribute clinical advice on my research plan. Caroline has always supported my clinical, managerial and professional opportunities. Caroline has collaborated with the CSP to promote the 'Making Activity Happen' and 'Love Activity, Hate Exercise' campaigns. She also added to the CSP's objective to advocate for the profession to key stakeholders and the general public through contribution to a BBC article on regaining strength after Lockdown. Caroline is a trustee of LEGS charity and a founding member of Long COVID Physio and which aims to provide peer support for healthcare workers living with Long COVID. Her breadth of roles and experiences in public sector, charitable and regulatory bodies will provide broad perspectives for my career development as a clinical academic.

Collaborations
Professor Bastian Bloem, Radboud UMC, Nijmegen, Netherlands. Professor Bloem is a world-leading expert on Parkinson's disease (PD). He founded ParkinsonNet, the world's foremost healthcare model for people with PD (PwP). Physiotherapy is important in his approach to research, and he has advocated the role of online exercise for PwP. After meeting at WPC 2023 Professor Bloem is keen to foster a collaboration with myself and support my studies. This will be facilitated by a visit in year 3. His expertise will also help development of the future RCT, consideration of measuring effectiveness through laboratory-based outcomes as well as someone to establish multi-centre site collaborations. Dr Rebecca Farley, Parkinson's Wellness Recovery PWR!, Tucson, Arizona. Dr Farley is a physiotherapist, neuroscientist, and PD exercise specialist. She is the Chief Scientific Officer and Founder of Parkinson Wellness Recovery PWR! Her current focus is the publication of research from her facility and integrating new research into workshops and her programmes. A visit will enable me to attend the PWR!Moves workshop as specified in my training & development plan and collaborate on developing translational approaches used by private sector/industry to clinical interventions and research within the NHS. Dr Farley and I have corresponded, and she is keen to facilitate a visit and explore future research collaboration of the PWR!Moves. Melissa McConaghy, Advance Rehab Centre, Sydney. Melissa is a Fellow of the Australian College of Physiotherapists and Specialist Neurological Physiotherapist. Melissa founded and curated the online global Parkinson's Conference, "INSIGHT into PD", and authored the book "The New Parkinson's Treatment: Exercise is

Medicine". She developed PD Warrior, an exercise intervention approach for treating Parkinson's disease. Melissa and I met at WPC 2023 and she welcomes a collaboration to discuss skills in engagement, influence and impact in disseminating my research and promoting myself as a clinical academic.

Neuro Heroes. Neuro Heroes is the UK's first online neuro gym, delivering condition specific exercise classes for PwP and other neurological conditions. They are keen for me to work with them in a supernumerary position in year 2 of my DCAF as part of my clinical practice to be undertaken. This collaboration will facilitate shared knowledge of running online exercise groups for PwP and enable me to build working relationships between the NHS, private sector industry and academia.

Clinical academic advisory group: I have established a network of peers with common research and clinical interests. The group utilises clinical experts in PD and exercise to guide development of the Power-PD project. The group includes:

Dr Julie Jones, Robert Gordon University (RGU), Aberdeen. Julie is a clinical academic fellow and a senior lecturer at RGU. She is a member of the Strategy Advisory board of Parkinson's UK in Scotland, chairs the Scotland wide Parkinson's Active project and is the co-lead of the Parkinson's Excellence Network Exercise hub. Julie completed her PhD exploring the feasibility of a collaborative approach to exercise delivery between the NHS and community sector.

Dr Bhanu Ramaswamy. Bhanu is an independent Physiotherapy Consultant and Honorary Visiting Fellow Sheffield Hallam University. Bhanu is the lead for the service user involvement committee for the Parkinson's Excellence Network.

Dr Anna Ferrusola Pastrana, University de Girona. Anna has been working in Parkinson's research since 2017 and has completed a PhD investigating the effects of exercise on people with Parkinson's disease at the University of Kent.

Ledia Alushi. Ledia is currently doing her PhD at the University of Cambridge to collaboratively develop and evaluate interventions that can be implemented to improve the physical, emotional and social well-being of people with Parkinson's Disease.

Host Organisation Support Statement

The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

James started working in the Camden Neurology and Stroke Service (CNSS) within Central and North West London NHS Trust in 2018. He has been a strong clinician bringing extensive experience of working in diverse clinical neurological settings as well as academic skills to the team. James has lead on service development projects including the development of an exercise group for Parkinson's disease (PD). James will have a supernumerary clinical role during the first year of his DCAF working 0.2fte. This will allow him to continue developing clinical and leadership skills through delivering exercise groups for and supporting people with PD (PwP). He will also provide clinical skills training, share expert knowledge and guidance for managing PD within the Trust.

James has demonstrated a clear career path with interest in continuing clinical work whilst also promoting and supporting research. As the Clinical Lead Physiotherapist with a PhD myself, I support James to take every opportunity to build on his clinical academic skills during the DCAF. CNWL trust is keen to develop research from Allied Health Professions and has close links with University College London (UCL). It is committed to equality of opportunity and believes that through its EDI mission statement it can achieve high quality evidence-based research and clinical care. James has met with Dr Kemp (Director of therapies), Johanne Watson (Chief Physiotherapist) and Prof Kilbride (AHP research lead) to agree to work together and develop clinical academic career opportunities within CNWL for those who have undertaken academic training. The clinical and research skills James will acquire will be of immense value to our team as we are keen to increase our research output and the North Central London Integrated Care System.

UCL boasts itself as an outstanding place to study and support research. It is ranked as one of the top 10 universities in the world (QS world rankings 2023). UCL was rated 2nd in the UK for research strength in the 2021

Research Excellence Framework. UCL's research environment achieved the maximum possible score, 100% at 4*. Specific to James' area of interest, UCL is the 3rd most highly cited university in Europe for neuroscience and behaviour studies (InCites scientific indicators 2018) and UCL biomedical and health researchers are the most influential in England, including the field of clinical neurology (RAND bibliometric analysis 2015). Moreover, our education, training and research is delivered in partnership with leading specialist hospitals with designated status as an Academic Health Sciences Centre. UCL prides itself on creating an inclusive culture for students and employees and received a silver Athena SWAN award in 2021.

Through UCL James will have access to world class resources, including a comprehensive doctoral training programme, 15 libraries and an excellent support network for research and academic writing. Doctoral students are supported via Thesis Committees throughout their PhD and report their progress on the Research Log. This method provides the student with wider intellectual contribution, access to copious expertise and an overview of the student's development at specific points during the degree. 'UCL Researcher Initiatives' supports early career researchers through a variety of learning, networking, and community listening opportunities.

James will be embedded academically in the UCL/Queen's Square Institute of Neurology (IoN), Department of Clinical and Movement Neurosciences. The department is a world-leading group bringing together expertise across the clinical and basic science spectrum of the physiology and pathology of human movement and movement disorders. The IoN has strong links with clinical service delivery working collaboratively with several NHS providers to promote the translation of research that is of direct clinical relevance into improved patient care and treatment. Boasting clinical and applied scientific activity the Institute is a unique national resource for postgraduate training and research in neurology, including James' interest of PD. Our mission is to provide the best quality learning and working experience, nurturing staff and students through being an inclusive and supportive workplace to reach their full potential. The institute supports clinical academic research careers through offering a wide range of training courses by the Graduate School and local department, as well as encouraging attendance at conferences and external meetings. The Institute offers a weekly teaching programme which James will be able to access as needed as a doctoral student. Students are given the opportunity to present their research, alongside their peers studying for PhDs, at the well-regarded annual Queen Square Symposium. James has also met with Dr Ramdharry who leads a research networking group for AHPs within the IoN. This will provide him with peer support during his fellowship from other AHPs in clinical and academic roles.

James' chosen area of research concerns the feasibility of a hybrid exercise programme for people with PD. James will be supervised by Prof Schrag who has extensive research experience in PD and clinical trials and alongside Professor Jones they will be able to support completion of his work packages and training & development plan. Students at the IoN are ideally placed to exploit the opportunities for career progression which arise from studying in a well-funded national and international centre of excellence. At UCL we are proud to be one of few institutions that has supported and funded clinical academic partnerships. Our intention is to develop the necessary skills, experience and attitudes for those who wish to pursue careers as clinical academics to do so. It is with great excitement that we can say we will support James to undertake a DCAF and supported by CNWL trust, foster a long-term partnership whereby a highly regarded academic institution is supporting clinically relevant research.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Shared Staff

Name	Rehabilitation Assistant

Employing organisation	University College London

Role on this research	Rehabilitation assistant

Grade	6B

Starting point on scale	25

Increment Date	

Justification of Costs
Agenda for change band 4 rehabilitation assistant to support running of intervention during WP2 (0.1wte for 18 months in total). Costing has been matched over to equivalent UCL pay scale for purpose of costing the project.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	32332
Geographical Weighting (£)	5000
Other allowances (£)	0
Superannuation (£)	853.8
Nat. Insurance (£)	597.13
Current Annual Costs (£)	**38782.93**

2

Staff type	Shared Staff

Name	Physiotherapist

Employing organisation	University College London

Role on this research	Physiotherapist

Grade	8

Starting point on scale	39

Increment Date	

Justification of Costs
Agenda for change band 6 physiotherapist to support running of intervention during WP2 (0.1wte for 18 months in total). Costing has been matched over to equivalent UCL pay scale for purpose of costing the project.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	48350
Geographical Weighting (£)	4500
Other allowances (£)	0
Superannuation (£)	1218.11
Nat. Insurance (£)	932.30
Current Annual Costs (£)	**55000.41**

3

Staff type	Lead Applicant

Name	James Alexander

Employing organisation	University College London

Grade	CL8

Starting point on scale	7

Increment Date	

Justification of Costs
Lead Fellow. Previously employed as a band 7 physiotherapist within the NHS. As of April 2025,beginning of fellowship, I would be on top pay step band 7. The equivalent salary has been mapped across to UCL non-consultant clinical academic pay scale and includes normal increments on that scale

Details of Posts and Salaries Costs

	Total (£)

	Total (£)
Salary (£)	50056
Geographical Weighting (£)	7746
Other allowances (£)	0
Superannuation (£)	8312
Nat. Insurance (£)	6721
Current Annual Costs (£)	**72835**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: Other
Other Type: Shared Staff
Name: **Research assistant - rehabilitation assistant for WP2**
Type of cost: HEI
% Full-time on this research: 10.0
Total months on this research: 18.0
Justification of Cost: Agenda for change band 4 rehabilitation assistant to support running of intervention during WP2 (0.1wte for 18 months in total). Costing has been matched over to equivalent UCL pay scale for purpose of costing the project.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	4739	2404	0	7143

Staff Type: Other
Other Type: Shared Staff
Name: **Senior research fellow - physiotherapist for WP2**
Type of cost: HEI
% Full-time on this research: 10.0
Total months on this research: 18.0
Justification of Cost: Agenda for change band 6 physiotherapist to support running of intervention during WP2 (0.1wte for 18 months in total). Costing has been matched over to equivalent UCL pay scale for purpose of costing the project.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	6805	3466	0	10271

Staff Type: (Lead Applicant)-Mr James Alexander
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: Previously employed as a band 7 physiotherapist within the NHS. As of April 2025,beginning of fellowship, i would be on top pay step band 7. The equivalent salary has been mapped across to UCL non-consultant clinical academic pay scale and includes normal increments on that scale

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	51558	53104	54698	159360
National Insurance	6923	7130	7344	21397
Superannuation	8561	8818	9083	26462

Geographical weighting / Other allowances (if applicable)	7978	8218	8464	24660

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	185500	0	13931	199431
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Subsistence
Conference: True
Item Description: Conference accommodation and subsistence
Justification of Cost: Attendance at national and international conferences is essential to build my network and disseminate my research for maximum impact. I have specifically chosen related conferences to attend each year and have spread the costs evenly over each conference and year up to the guidance budget limit. Conferences planned to attend include: Year 2: ACPIN conference -UK. World Parkinson's congress (WPC) 2026 – Phoenix, USA. Year 3: WCPT - TBC

	Year 1	Year 2	Year 3	Total Cost (£)
	0	450	450	900

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference fee
Justification of Cost: Attendance at national and international conferences is essential to build my network and disseminate my research for maximum impact. I have specifically chosen related conferences to attend each year and have spread the costs evenly over each conference and year up to the guidance budget limit. Conferences planned to attend include: Year 2: World Parkinson's congress (WPC) 2026 Phoenix, USA. ACPIN conference - UK. Year 3: WCPT - TBC

	Year 1	Year 2	Year 3	Total Cost (£)
	0	550	350	900

Type of cost: HEI
Description: Journey Costs
Conference: True
Item Description: Conference travel
Justification of Cost: Attendance at national and international conferences is essential to build my network and disseminate my research for maximum impact. I have specifically chosen related conferences to attend each year and have spread the costs evenly over each conference and year up to the guidance budget limit. Conferences planned to attend include: Year 2: World Parkinson's congress (WPC) 2026 – Phoenix, USA. ACPIN conference - UK. Year 3: WCPT - TBC

	Year 1	Year 2	Year 3	Total Cost (£)
	0	600	600	1200

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Fellow international accommodation and subsistence
Justification of Cost: Accommodation and subsistence for international visits in line with UCL expenses policy. Calculation based on recent internet searches for nearby accommodation to training/site and subsistence

calculated at £37/day. Year 2 visit to Dr Rebecca Farley at PWR! Parkinson Wellness Recovery, Arizona, USA as part of training and development and collaborations. 5x nights (£150/night) and subsistence (£37/day) = £935. Year 3 visit to Prof Bloem at Radboud UMC, Nijmegen, Netherlands as part of collaboration visit. 3x nights (£150/night) and subsistence (£37/day) = £561. Total

	Year 1	Year 2	Year 3	Total Cost (£)
	0	935	561	1496

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Fellow international travel
Justification of Cost: All travel to be booked through UCL approved provider and in line with UCL travel policy. International travel includes: Year 2. Return flights to Tuscon, Arizona USA for collaboration visit with Dr Becky Farley and to attend PWR!Moves training. In line with UCL Expenses policy, a flight greater than 5 hours in duration qualifies for "premium economy" costs. Based on current internet search this is costed at £1,500. Year 3. Economy return train travel from London to Nijmegen for collaboration visit with Professor Bloem costed at £200.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1500	200	1700

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Fellow UK accommodation and subsistence
Justification of Cost: Accommodation and subsistence for international visits in line with UCL expenses policy. Calculation based on recent internet searches for nearby accommodation to training/site and subsistence calculated at £37/day. Year 1: Queen Mary feasibility workshop, Oxford, accommodation 1x (£150/night) plus subsistence 1x (£37/day) = £187. Uni. Salford Parkinson's course, Salford accommodation 3x (£120/night) plus subsistence 3x (£37/day) = £471. Year total = £658. Year 2: Oxford Uni RCT course, Oxford, accommodation 5x (£120/night) plus subsistence 5x (£37/day) = £785. Atypical Parkinson's course, Derby. Accommodation 1x (£120/night) plus subsistence 1x (£37/day) = £157. Year total = £942.

	Year 1	Year 2	Year 3	Total Cost (£)
	658	942	0	1600

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Fellow UK travel
Justification of Cost: To be booked through UCL approved provider and in line with UCL travel policy. Economy rail. Includes: Year 1: Feasibility study workshop Oxford University, 1 x return rail journey = £80. University of Salford Parkinson's course 3x return rail journey 3x £133 = £399. Year 2: RCT course Oxford University. 1x return rail journey = £80. Atypical Parkinson's course, Derby. 1x Return rail journey = £80. Total = £160 Travel costs to central London (Zone 1-6 annual travelcard) for clinical practice roles during DCAF, all related travel within London for WP1 and WP2 (including observation visits, data collection, interviews) and all travel for training and development within London (KCL implementation course). Based on current Transport for London fees - £2,976 per year

	Year 1	Year 2	Year 3	Total Cost (£)
	3455	3136	2976	9567

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Intervention participant travel costs
Justification of Cost: Participant travel costs for travel to intervention venue for WP2 based on Transport for London zone 1-3 off-peak travel charge = £5.40 return (£5.40 x 12 weeks) x50 participants = £3,240

	Year 1	Year 2	Year 3	Total Cost (£)
	0	2592	648	3240

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: PPI accommodation and subsistence
Justification of Cost: 3x Subsistence for in-person workshops (£10/day x6 participants = £180 (£60 for each year of PhD)). 3x overnight accommodation for attendance of in-person workshop x6 participants (£100 per person = £1,800 total (£600 each year of PhD)). Total = £1980

	Year 1	Year 2	Year 3	Total Cost (£)
	660	660	660	1980

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: PPI travel costs
Justification of Cost: Travel expenses for 3x in-person workshops (£50p/person x6 participants = £900 total (£300 each year of PhD)).

	Year 1	Year 2	Year 3	Total Cost (£)
	300	300	300	900

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	5073	11665	6745	23483

Equipment
No costs have been added

Consumables

Description: Consumables (food and drink) for peer support session WP2
Type of cost: HEI
Please describe fully: Consumables (food and drink) for peer support session WP2
Justification of Cost: Consumables for peer support session WP2 e.g. tea, coffee, snacks, cups, plates etc.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	125	125	250

Description: GENEActiv accelerometer charger x 12
Type of cost: HEI
Please describe fully: GENEActiv accelerometer charger x 12
Justification of Cost: Accelerometers to measure physical activity levels during intervention period of WP2. A quote was received from ActivInsights company who produce the GENEActiv accelerometer and chargers. Prices for the chargers are £160.00 + VAT. 12 will be required for max participants per exercise group (12x £160 - £1,920 + VAT)

	Year 1	Year 2	Year 3	Total Cost (£)
	2320	0	0	2320

Description: GENEActiv accelerometer x 12
Type of cost: HEI
Please describe fully: GENEActiv accelerometer x 12
Justification of Cost: Accelerometers to measure physical activity levels during intervention period of WP2. A quote was received from ActivInsights company who produce the GENEActiv accelerometer and chargers. Prices for the accelerometers are £250.00 + VAT. 12 will be required for max participants per exercise group (12x £250 - £3,000 + VAT)

	Year 1	Year 2	Year 3	Total Cost (£)
	3600	0	0	3600

Description: Research textbooks
Type of cost: HEI
Please describe fully: Research textbooks
Justification of Cost: Research textbooks (statistics, quantitative research methods, implementation science, designing clinical research) £100 each

	Year 1	Year 2	Year 3	Total Cost (£)
	400	0	0	400

Description: Encrypted audio recording device
Type of cost: HEI
Please describe fully: Encrypted audio recording device
Justification of Cost: Required for sound recording face to face qualitative interviews. Based on internet searches, cost = £450

	Year 1	Year 2	Year 3	Total Cost (£)
	450	0	0	450

Description: Exercise equipment for intervention during WP2
Type of cost: HEI
Please describe fully: Exercise equipment for intervention during WP2
Justification of Cost: Exercise equipment required for intervention in WP2 including theraband (£200), PD warrior equipment (£200), dumbbell weights (£200), exercise mats (£170), pilates blocks (£100), foam balls (£130)

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	0	0	1000

Description: Laptop for home working, remote working & fieldwork
Type of cost: HEI
Please describe fully: Laptop for home working, remote working & fieldwork
Justification of Cost: Laptop (excluding VAT) for home working, remote working & fieldwork. Approximately £1,000 based on internet searches

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	0	0	1000

Description: Laptop for use during WP2
Type of cost: HEI
Please describe fully: Laptop for use during WP2
Justification of Cost: Laptop (excluding VAT) for use during intervention in WP2 for instructors to use and to enable participants to join online. Based of current internet search approximately £1,000

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	0	0	1000

Description: Technology equipment for WP2
Type of cost: HEI
Please describe fully: Technology equipment for WP2
Justification of Cost: Technology equipment for WP2 to run hybrid exercise intervention. Costs have been calculated from internet search for equipment: wide angle web camera (£90), tripod (£25), projector (£160), projector screen (£80), Ringlight (£80), wireless microphone system (£50), board and pens (£30).

	Year 1	Year 2	Year 3	Total Cost (£)
	515	0	0	515

Description: Non-exercise equipment for WP2
Type of cost: HEI
Please describe fully: Non-exercise equipment for WP2
Justification of Cost: Non-exercise equipment required for intervention in WP2 and to complete outcome measures e.g. 2x stopwatch (£10/each = £20), tape measure (£10), music speakers (£80), pens (£10), clipboards x 12 (£30).

	Year 1	Year 2	Year 3	Total Cost (£)

	150	0	0	150

Description: Printing and postage during project and dissemination
Type of cost: HEI
Please describe fully: Printing and postage during project and dissemination
Justification of Cost: Printing and paper during years 1-3 for WP2 (consent form, PIS, outcome measures, questionnaires, participant exercise booklet) 2000 pages @5p = £100. Postage costs (recruitment materials, questionnaires, incentives) 200x £2 = £400. Consumables associated with dissemination. 2x conference poster printing = £100. Infographic printing = £200. Total printing and postage = £800+ VAT = £960.

	Year 1	Year 2	Year 3	Total Cost (£)
	240	360	360	960

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	10675	485	485	11645

Patient and Public Involvement

Description: PPI payments
Type of cost: HEI
Please describe fully: PPI payments
Justification of Cost: All cost for PPI and advisory group involvement based on NIHR “Budgeting for involvement: Practical advice on budgeting for actively involving the public in research studies”. Over the 3 year project the following PPI will be included: PPI - 6x online meetings (2hrs x6 participants @ £50/h per person = £3,600 (£1,200 each year of PhD)). 3x in-person workshops (£300 p/day x6 participants = £5,400 (£1,800 each year of PhD)). Clinical academic advisory group: 8x online group meetings (1hr x5 participants @£150 each per meeting = £6,000 total over duration of PhD).

	Year 1	Year 2	Year 3	Total Cost (£)
	5250	5250	4500	15000

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	5250	5250	4500	15000

Other Direct Costs

Description: Intervention participant incentives
Type of cost: HEI
Please describe fully: Intervention participant incentives
Justification of Cost: Participation incentives for WP2. £10x 100 non-participants for TFA questionnaire = £1,000. £20x 50 for participation in WP2 intervention = £1,000. £10x 50 WP2 participant questionnaire & interview = £500. Total = £2,500.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1250	1250	2500

Description: Venue hire
Type of cost: HEI
Please describe fully: Venue hire
Justification of Cost: The primary plan is to utilise an NHS rehabilitation gym within UCLH or CNWL recently therapy gym spaces have been threatened to be taken away due to budget cuts. Therefore to ensure the provision of a site to conduct WP2 i have costed in the hire of a suitable space. It is hoped this will not be needed but felt important to do to not impede the continuation of the project should NHS gym space be not available. Hiring of venue for delivery of intervention for WP2. Based on internet search of costs of hiring community venues. Venue hire (£120 p/hour x 3hrs = £360 x 64 weeks (duration of WP2 intervention in total = £23,040)

	Year 1	Year 2	Year 3	Total Cost (£)
	1440	14400	7200	23040

Description: Software licences
Type of cost: HEI
Please describe fully: Software licences
Justification of Cost: Software licence for Microsoft Project

	Year 1	Year 2	Year 3	Total Cost (£)
	360	0	0	360

Description: Transcription costs
Type of cost: HEI
Please describe fully: Transcription costs
Justification of Cost: Transcription of WP2 interviews (Transcription through professional provider costed at £0.96/min + VAT x 60min interviews x 50 participants = £3,444)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1722	1722	3444

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	1800	17372	10172	29344

Training And Development

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: PhD tuition fees
Justification of Cost: I am applying to undertake this PhD full time. UCL annual full time fees for 2025/26 are £6,215 per year. Total = £18,645

	Year 1	Year 2	Year 3	Total Cost (£)
	6215	6215	6215	18645

Type of cost: HEI
Description: Short Courses
Item Description: Design & implementation of digital interventions course
Justification of Cost: Course fees based on completing the 30 hour online course in 2 months at £38 per month

	Year 1	Year 2	Year 3	Total Cost (£)
	76	0	0	76

Type of cost: HEI
Description: Short Courses
Item Description: Developing & testing complex interventions course
Justification of Cost: Course fees based on current cost

	Year 1	Year 2	Year 3	Total Cost (£)
	99	0	0	99

Type of cost: HEI
Description: Workshops
Item Description: Joanna Briggs Institute scoping review workshop
Justification of Cost: Course fees for JBI scoping review workshop based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	400	0	0	400

Type of cost: HEI
Description: Short Courses
Item Description: King's Fund Emerging Clinical Leaders course
Justification of Cost: Course fees for King's fund emerging leaders course based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	2700	2700

Type of cost: HEI
Description: Short Courses
Item Description: NCORE Atypical Parkinson's course
Justification of Cost: Course fees for NCORE atypical Parkinson's course based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	0	150	0	150

Type of cost: HEI
Description: Short Courses
Item Description: Oxford University RCT course
Justification of Cost: Course fees for Oxford Uni RCT course based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1550	0	1550

Type of cost: HEI
Description: Workshops
Item Description: PWR!Moves workshop
Justification of Cost: Course fees for PWR!Moves workshop based on current in-person fees

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	550	550

Type of cost: HEI
Description: Workshops
Item Description: Queen Mary Pilot & Feasibility study workshop
Justification of Cost: Course fees for QMU pilot & feasibility study workshop based on previous fees

	Year 1	Year 2	Year 3	Total Cost (£)
	200	0	0	200

Type of cost: HEI
Description: Short Courses
Item Description: UCL Understanding health economics in clinical trials course
Justification of Cost: Course fees based on current UCL Understanding health economics in clinical trials course course fees

	Year 1	Year 2	Year 3	Total Cost (£)
	135	0	0	135

Type of cost: HEI
Description: Short Courses
Item Description: University of Salford Parkinson's disease course
Justification of Cost: Course fees for University of Salford Parkinson's disease module based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	1500	0	0	1500

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	8625	7915	9465	26005

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	96113	102565	90937	289615

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: UCL
Justification of Cost: Estate costs

	Year 1	Year 2	Year 3	Total Cost (£)
	21725	19914	18104	59743

Type of cost: Other Indirect Costs
Name of Higher Education Institute: UCL
Justification of Cost: Infrastructure Technician Costs

	Year 1	Year 2	Year 3	Total Cost (£)
	1565	1434	1304	4303

Type of cost: Other Indirect Costs
Name of Higher Education Institute: UCL
Justification of Cost: Indirect Costs

	Year 1	Year 2	Year 3	Total Cost (£)
	82407	75539	68672	226618

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	105697	96887	88080	290664

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	84558	77509	70464	232531

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	260610	232531	493141
Total Higher Education Institution Costs (100%)s	29005	0	29005
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0

Total NHS Research Costs	0	0	0
Total Research Costs Requested From Funder	289615	232531	522146

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**69250**	**66511**	**63670**	**199431**
Travel-subsistence-and-conference-fees	**4058**	**9653**	**5676**	**19387**
Equipment	**0**	**0**	**0**	**0**
Consumables	**8540**	**388**	**388**	**9316**
Patient-and-Public-Involvement	**4200**	**4200**	**3600**	**12000**
Other-Direct-Costs	**1440**	**13898**	**8138**	**23476**
Training And Development	**8625**	**7915**	**9465**	**26005**
HEI-Indirect-costs	**84558**	**77509**	**70464**	**232531**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**180671**	**180074**	**161401**	**522146**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
NHS service support cost from SoECAT	0	55	55	110.0

Justification:
NHS service support cost from SoECAT: NHS service support cost derived from SoECAT for WP2. This was completed in collaboration with the UCLH joint research office and LCRN North Thames. Total = £110.08. I have distributed these as expected over years 2 & 3 years of the DCAF.

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Support Costs Total	0	55	55	110

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
NHS treatment costs from SoECAT	0	0	0	0.0

Justification:
NHS treatment costs from SoECAT: NHS treatment costs derived from SoECAT. This was completed in collaboration with the UCLH joint research office and LCRN North Thames. The intervention during WP2 being investigated is in addition to usual care therefore there are no excess NHS treatment costs.

	Year 1	Year 2	Year 3	Total Cost (£)

NHS Treatment Costs Total	0	0	0	0

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
N/A

Usual Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
Usual treatment costs associated with WP2	0	0	0	0.0

Justification:
Usual treatment costs associated with WP2: There is no standardised usual care. The intervention during WP2 under investigation is in addition to usual care.

	Year 1	Year 2	Year 3	Total Cost (£)
Usual Treatment Costs Total	0	0	0	0

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	110
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	110

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	522146
Total NHS (Support and Excess Treatment) Costs	110
Total Funding Required	522256

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as an Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - DCAF references 2024.docx) is included as an appendix within this file.

(Figures / Tables - DCAF 2024 upload figures .docx) is included as an appendix within this file.

(Research Timetable - DCAF 2024 GANTT chart.docx) is included as an appendix within this file.

(SoECAT Template - Funder Export_329635_SoECAT_Power-PD_ EmPOWERing people with PD through flexible exercise_.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Participants
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required participant. Everyone included in this section will be acting as a 'participant' in your application and must agree to be named as such.

By confirming participation, participants are acknowledging their involvement and input into this application, and are agreeing to be involved, as indicated, in any resultant fellowship.

Once the 'Administrative Authority or Finance Officer' participant has confirmed their participation, they will be able to access the 'Detailed Budget' section of the application form. You should collaborate with your 'Administrative Authority or Finance Officer participant to ensure that this section is completed satisfactorily.

You must ensure that all participants are happy for your application to be submitted before you submit it.

It is permissible for a single individual to serve as both a signatory and a participant, but they will need to be added separately under each heading.

Named participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please refer to the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Professor
Forename(s)	Anette
Surname	Schrag
Position	Professor

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Professor
Forename(s)	Fiona
Surname	Jones
Position	Professor Rehabilitation Research

Practice Supervisor(s)

Practice Support Participant 1

Title	Dr
Forename(s)	caroline
Surname	appel
Position	Clinical Lead Physiotherapist

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mrs
Forename(s)	Kasia
Surname	Haremza
Position	Assistant Director of Research Development and Support (Pre-Award)

Signatories
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required 'Head of Department / Senior Manager' signatory. Once their contact details have been entered below, these signatories can be invited to log in and confirm their participation in your application.

Once the nominated signatories have confirmed their participation, they can access and complete the required 'Host Organisation Statement of Support' in the 'Training & Development and Research Support' section of the application form.

Your signatories must confirm their participation in your application and complete the section they are responsible for before you can select the 'submit' option.

Signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

Signatory approval from your 'Head of Department / Senior Manager' signatories will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system-generated email.

The NIHR will not accept your application unless fully approved by your signatories before the 1 p.m. deadline on Tuesday 11th June 2024.

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Dr

Forename(s)	caroline
Surname	appel
Position	Clinical Lead Physiotherapist

Head of Department or Senior Manager 2

Title	Professor
Forename(s)	Anthony
Surname	Schapira
Position	

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
There are no conflicts to declare

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

Appendices

1. DCAF references 2024.docx
2. DCAF 2024 upload figures .docx
3. DCAF 2024 GANTT chart.docx
4. Funder Export_329635_SoECAT_Power-PD_ EmPOWERing people with PD through flexible exercise_.xlsx

Appendix #1: DCAF references 2024.docx

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Appendix #2: DCAF 2024 upload figures .docx

Figure 1. Flow diagram illustrating WP2 study design and data collection points

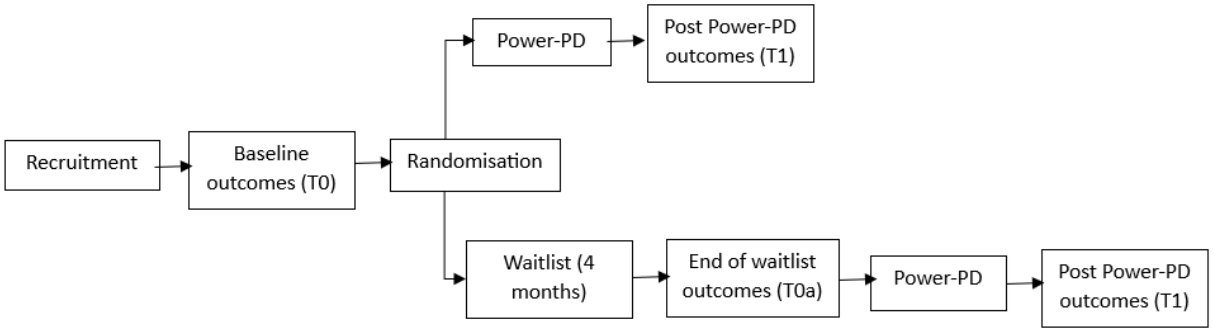
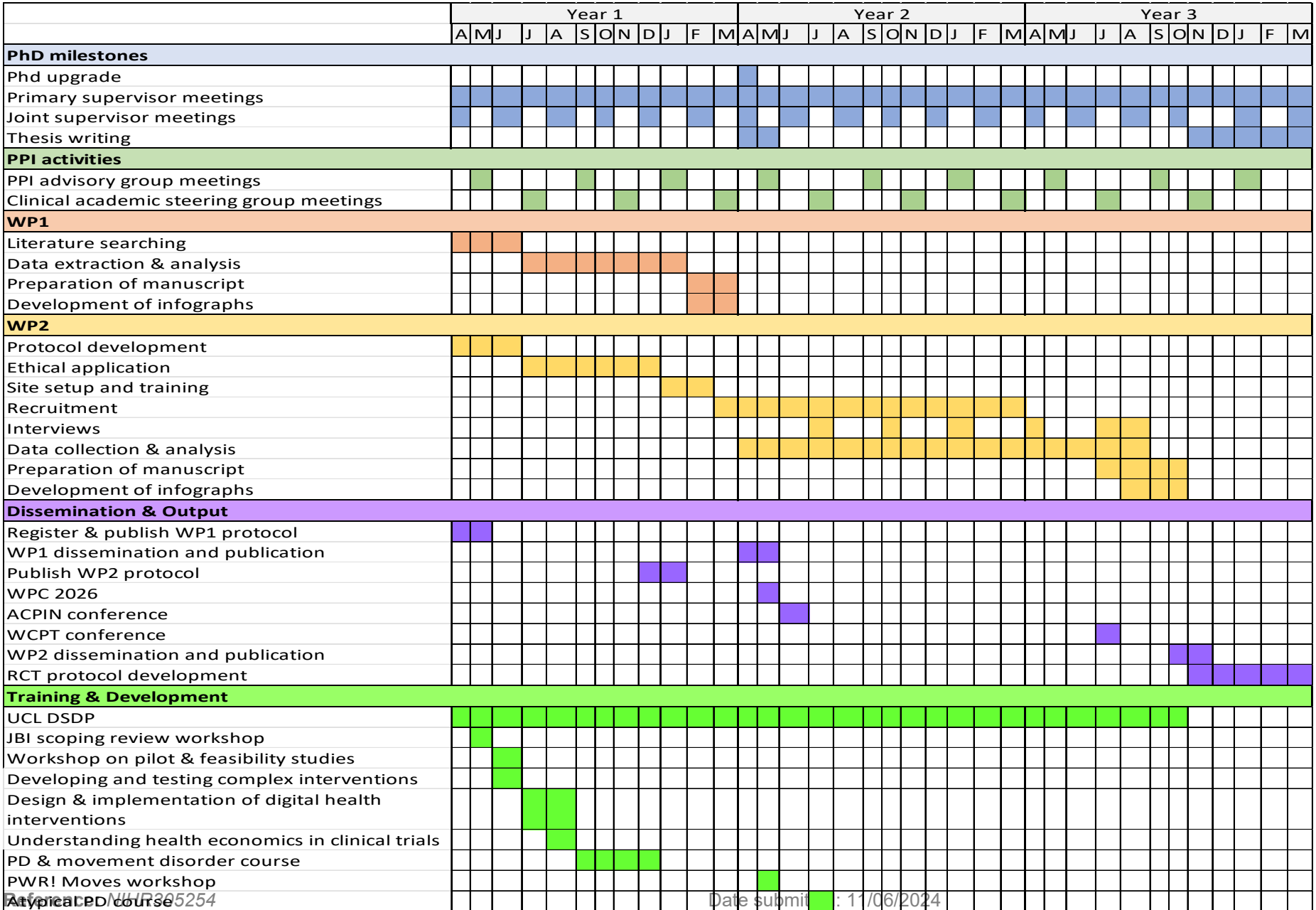


Table 1. Table illustrating outcome measures to be collectedrelevant to project aims and objectives.

Aim	Objective	Measures
Acceptability	Representative recruitment and engagement Adherence to study Acceptability to be randomised	**Quantitative:** Demographic data (age, socioeconomic status, education level and ethnicity) Prospective acceptability through TFA questionnaire Adherence & attrition rates **Qualitative:** TFA questionnaire Interviews with participants and practitioners
Feasibility	Time to collect, clean and analyse data	No. of completed data sets Recruitment rates Retention rates
Fidelity	Practicality of carrying out intervention	**Quantitative:** Fidelity measure checklist **Qualitative:** Interviews with participants and practitioners Observation
Secondary Outcomes		
Efficacy	Choice of outcome measures	**Physical activity levels** PA diaries Body worn accelerometer **Physical/motor function** Timed up and go (TUG) **Self-efficacy** Self-efficacy for exercise scale (SEE) Importance/Confidence scale
Cost benefit outcomes		
	Health economic outcomes	EQ-5D-5L PDQ-8
	Resource use	Client service user inventory

Appendix #3: DCAF 2024 GANTT chart.docx


Year 1
Year 2
Year 3
A M J J A S O N D J F M A M J J A S O N D J F M A M J J A S O N D J F M
PhD milestones
Phd upgrade
Primary supervisor meetings
Joint supervisor meetings
Thesis writing
PPI activities
PPI advisory group meetings
Clinical academic steering group meetings
WP1
Literature searching
Data extraction & analysis
Preparation of manuscript
Development of infographs
WP2
Protocol development
Ethical application
Site setup and training
Recruitment
Interviews
Data collection & analysis
Preparation of manuscript
Development of infographs
Dissemination & Output
Register & publish WP1 protocol
WP1 dissemination and publication
Publish WP2 protocol
WPC 2026
ACPIN conference
WCPT conference
WP2 dissemination and publication
RCT protocol development
Training & Development
UCL DSDP
JBI scoping review workshop
Workshop on pilot & feasibility studies
Developing and testing complex interventions
Design & implementation of digital health interventions
Understanding health economics in clinical trials
PD & movement disorder course
PWR! Moves workshop
Atypical PD course
Oxford RCT course
Parkinson's train the trainer course
Emerging leaders course
Clinical Activities
CNSS

Appendix #4: Funder Export_329635_SoECAT_Power-PD_ EmPOWERing people with PD through flexible exercise_.xlsx